Establishment of Erickson Tract *Poria* Control Plots

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Preliminary work: At the invitation of John Hammons, Division Forester, International Paper Co. (IPCo), I visited *Poria* infested company lands in Oregon and Washington. Two potential study areas in Columbia Co., Oregon were selected. Prof. Lewis Roth, OSU, Dr. James Trappe, PNW Experiment Station, John Hammons, IPCo, Earl Nelson and I examined the areas and tentatively agreed to develop a cooperative research program. In early December, Dr. Nelson and I, and one technician ran a preliminary 5% survey on the two stands to assess the uniformity and intensity of infection.

Poria killed trees were mapped on strips one chain wide and five chains apart through the stands. This took a total of seven days in record low temperatures, snow and sleet. Larry Weir, Forest Pathologist, Oregon State Department of Forestry, accompanied us during the worst weather. As a result of the survey, a 35 acre stand of 40 year old Douglas fir near the middle of section 17, T6N, R3W, known as the Erickson Tract, was selected for further development. The stand lies on a NE slope at about 1800' elevation near Apiary, Oregon, 15 miles from Longview, Washington, and 3 hours drive from Corvallis.

Plot establishment: On March 9, 1973, representatives of IPCo., Crown Zellerbach, Oregon State University, USFS, and the Oregon State Department of Forestry met in Longview to develop working plans for the cooperative program. IPCo accepted a research proposal to test red alder as a biological

control for Poria, and agreed to perform the logging, regeneration, and protection phases of the study, as well as provide $5,000 to support the research program. The Forest Service agreed to develop a study plan, lay out plots, and collect data for the duration of the study. It was agreed that additional industrial cooperators would be sought to support this research and provide additional study areas. In the summer of 1973 Crown Zellerbach officially joined the cooperators and provided $2,000 for research support. The role of Dr. Ralph Shay, at that time Head of the Department of Botany and Plant Pathology at Oregon State University, should be emphasized. Dr. Shay was involved in the initial contacts with IPCo., and subsequently aided and encouraged my work in the project.

In March and April, a total of 15 days with a three or four man crew were spent on plot layout and mapping. A grid was established over the area, with 24" steel stakes driven 20" into the ground at 2 chain (132') intervals. An aluminum flasher stapled to the base of a nearby tree marked each stake. Each flasher was scribed with the number of and distance to the stake, and faced toward the stake. A control line was first run for 34 chains through the middle of the plot on a bearing of 135° (S45oE). A second control line was run at right angles to the first 225°, N45°E), at the 14 chain station. Control lines were run with staff compass using backsight readings, and steel tape. The declination was 22°, and all distances were corrected for slope. The other lines were located in a similar manner. The stakes are identified according to their position on the primary control line (0 to 34), and their distance in chains up slope or down slope from the primary line (+12 to C to -10). Thus the stake at the junction of the control lines is 14C, while the stakes two chains away are labeled 14+2, 16C, 14-2, and 12C. The grid blocks are

numbered from left to right (SE to NW) starting in the E corner of the plot. The grid lines were permanently referenced beyond the edges of the plot to allow reestablishment after site disturbance. Because of stand conditions and Poria distribution, the plot boundary was to be the ridge above and the road below, and the 8 line on the NW and the 34 line on the SE.

After grid establishment, each block was examined for evidence of Poria. All dead and dying trees were examined and those with Poria recorded by azimuth and distance from the compass point, which was in turn referenced to the corners of the grid block. Thirty-four acres were systematically examined for Poria. Poria killed trees were present on 68 of the 85 grid blocks examined. The location of each Poria killed tree was marked on a map by a symbol representing its condition.

The tree data are also recorded on data punch cards, and a computer program has been developed by Robert Denison that sorts the trees by location, and generates a computer map of the area. If the program proves flexible enough, it will be of great use in future data analysis. Poria infected trees are identified on the computer printout and map according to block number, and x, y coordinates in 2-chain units from 0,0 (34-10 on the hand drawn map). Thus the stakes around the intersection of the control lines, 14C on the hand drawn map, and 10,5 on the computer map, are: 10,6 = 14+2, 9,5 = 16C, 10,4 = 14-2, and 11,5 = 12C. A summary of the program's contents and capabilities is included with the data cards and printout.

After logging, corners were reestablished and the area was systematically examined for Poria infected living trees by checking the stumps for decay or stain. The location of infected stumps was recorded by grid blocks as before,

and this information added to the maps and data cards. The flashers were very helpful in relocating stakes, but some flashers were damaged. The bearing and distance from flasher to stake should have been recorded in the field notes.

Before logging, a soil survey was completed by George Otte of the Soil Conservation Service. Soil cores were taken at each of the grid corners. The resulting map identified three different soils, Hembre, Tolke, and Klickitat, with several slope phases. Soil samples were later collected systematically over the area by K. C. Lu, C. Y. Li, and I for nitrogen analysis and microbiological assay. This will serve as base information, so that changes after treatment can be followed.

The stand was systematically photographed before logging to record visually the intensity of Poria infection in each grid block. The 33 mm color slides and accompanying notes are part of the plot data record. It was hoped to obtain large scale aerial photography of the area to supplement the record, but arrangements with John Wear of Insect and Disease Control, USFS, fell through.

The original schedule agreed to by the cooperators called for IPCo. to harvest the study area during the summer of 1973, so it could be burned that fall, and planted that winter. An access road was constructed through the southeast end of the plot and along the upper ridge. Although IPCO informed us of their road building plans, the work was finished before we could suggest a detailed routing. We consequently lost some valuable Poria infected ground from our study area. IPCo. was unable to contract a logger until late September, when they moved a thinning operation already under contact down to the Erickson Tract. Due primarily to

difficulties in keeping fallers on the job, the work progressed slowly and was not completed after 5 weeks. Slash burning was completed in the Spring of 1974. The formal study plan contains further details about plot establishment.